PARENT ONLY — SOLVED ANSWER KEY

Vyom · Parent answer key

26 Oct – 1 Nov 2026 · Monday–Sunday · NDPS Class 9

Vyom ko test se pehle ye PDF mat dena. Student pack mein sirf questions aur blank answer space hai. Is alag parent PDF mein har original question ke saath uska matching solved answer, working aur marking points diye hain; Test ID se student paper match karein.

Checking: same evening. C = concept, R = recall, K = careless, T = time.

90%+: short revision. 75–89%: weak-topic practice + retest. Below 75%: topic dobara samjhao; mistakes notebook mein likho.

Answer keys existing source question banks se li gayi hain; school pattern abhi unconfirmed hai. Alternate valid answers ko marking guidance ke mutabik accept karein.

Week score register

Date / test ID	Test	Marks	Galti / retest
26 Oct T2026-10-26	Science Weekly test	___ / 40	Retest: 29 Oct
28 Oct D2026-10-28	Daily mixed Daily test	___ / 10	Retest: 31 Oct
29 Oct D2026-10-29	Daily mixed Daily test	___ / 10	Retest: 1 Nov
30 Oct D2026-10-30	Daily mixed Daily test	___ / 10	Retest: 2 Nov
31 Oct D2026-10-31	Daily mixed Daily test	___ / 10	Retest: 3 Nov
1 Nov D2026-11-01	Daily mixed Daily test	___ / 10	Retest: 4 Nov

Science · Weekly Test 1 · Answer key

Monday, 26 October 2026 · 40 marks

Test ID: T2026-10-26

Class 9 · Ch 1 Exploration · Ch 2 Cell · Ch 3 Tissues in Action · Ch 8 Journey Inside the Atom (8.1–8.6)

Q1 [1]

Question: Meristematic cells can divide repeatedly because they have:

(a) thick walls for protection

(b) large vacuoles that store nutrients

(c) thin walls, dense cytoplasm and a large prominent nucleus

(d) functionally differentiated cells

Expected answer / marking points: (c)

Q2 [1]

Question: If a plant cannot transport food from its leaves to its roots, which tissue is not working properly?

(a) Xylem (b) Phloem

(c) Epidermis (d) Sclerenchyma

Expected answer / marking points: (b) Phloem

Q3 [1]

Question: Epithelial tissues lining internal organs are usually only one or a few cells thick in order to:

(a) store food (b) give maximum strength

(c) allow quick exchange of materials (d) reduce friction

Expected answer / marking points: (c) quick exchange of materials

Q4 [1]

Question: Which type of joint works when you bend your knee?

(a) Ball and socket (b) Hinge

(c) Pivot (d) Fixed

Expected answer / marking points: (b) Hinge

Q5 [1]

Question: The organelle that packs and stores materials received from the endoplasmic reticulum is:

(a) Lysosome (b) Ribosome

(c) Golgi apparatus (d) Vacuole

Expected answer / marking points: (c) Golgi apparatus

Q6 [1]

Question: In Rutherford's gold-foil experiment, most alpha particles passed straight through the foil. This showed that:

(a) the atom is mostly empty space

(b) electrons are heavier than protons

(c) the nucleus is negatively charged

(d) the atom is a solid ball

Expected answer / marking points: (a) mostly empty space

Q7 [1] **Question:** An atom has 79 protons and mass number 197. The number of neutrons is:

(a) 79 (b) 118

(c) 197 (d) 276

Expected answer / marking points: (b) 197 – 79 = 118

Q8 [1] **Question:** The correct historical order of atomic models is:

(a) Thomson → Dalton → Bohr → Rutherford

(b) Dalton → Thomson → Rutherford → Bohr

(c) Dalton → Rutherford → Thomson → Bohr

(d) Rutherford → Thomson → Dalton → Bohr

Expected answer / marking points: (b) Dalton → Thomson → Rutherford → Bohr

Q9 [1] **Question:** In science, a **theory** is:

(a) a guess that has not been tested

(b) an explanation of why patterns occur, based on evidence and careful testing

(c) a law that can never change

(d) a measurement written in SI units

Expected answer / marking points: (b)

Q10 [1] **Question: A:** Cardiac muscle can contract continuously without fatigue.
R: Cardiac muscle cells have many mitochondria and an abundant blood supply.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (a) Both true, R explains A.

Q11 [1] **Question: A:** Tendons connect bone to bone and allow joint movement.
R: Tendons are made of tough connective tissue that transmits force from muscle to bone.

(a) Both A and R are true, and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true, but R is false.

(d) A is false, but R is true.

Expected answer / marking points: (d) A false (tendons connect muscle to bone; ligaments connect bone to bone); R true.

Q12 [2] **Question:** Differentiate between rough endoplasmic reticulum (RER) and smooth endoplasmic reticulum (SER) on the basis of structure and function.

Expected answer / marking points: RER has ribosomes on its surface (rough look) and helps make proteins [1]; SER has no ribosomes (smooth) and helps make lipids [1].

Q13 **Question:** A young mango sapling bends in strong monsoon winds but does not break. Name the tissue responsible and predict what would happen if it were replaced by sclerenchyma. **[2]**

Expected answer / marking points: Collenchyma — living cells with walls thickened at the corners, gives flexibility with support [1]. Sclerenchyma has thick, rigid, lignified dead cells, so the stem would become stiff and could snap in strong wind [1].

Q14 **Question:** A magnesium atom has mass number 24 and atomic number 12. Find the number of (i) protons (ii) neutrons (iii) electrons. Show how you got the neutrons. **[2]**

Expected answer / marking points: Protons = 12, electrons = 12 (neutral atom) [1]; neutrons = mass number – atomic number = 24 – 12 = 12 [1].

Q15 **Question:** You want to make a simple scientific model to predict whether a cricket ball hit for a six will cross the boundary. Name two details you would include and two you would ignore. **[2]**

Expected answer / marking points: Include: mass of ball, speed and direction of hit (any two) [1]. Ignore: brand of bat, colour of ball, grass on field, seam stitching (any two) [1].

Q16 **Question:** Mitochondria and chloroplasts are two important organelles of a plant cell. Give two similarities and one difference between them. **[3]**

Expected answer / marking points: Similarities: both have a double membrane; both have their own DNA (and ribosomes); both are involved in energy conversion (any two) [2]. Difference: mitochondria release energy by respiration and are found in plant and animal cells; chloroplasts make food by photosynthesis and are found only in green plant cells [1].

Q17 **Question:** Coconut husk fibres are tough and used to make mats. Which plant tissue gives them this strength? Describe its structure, and explain why living parenchyma cannot do the same job. **[3]**

Expected answer / marking points: Sclerenchyma [1]. Cells are dead, long and narrow with thick lignified walls and little or no space inside [1]. Parenchyma cells are living, thin-walled and loosely packed, so they cannot provide toughness [1].

Q18 **Question:** Write any two main ideas of Bohr's model of the atom. Why did Rutherford's model fail to explain the stability of the atom? **[3]**

Expected answer / marking points: Bohr: electrons revolve only in certain fixed orbits/shells (K, L, M, N) with fixed energy; while in an orbit an electron does not lose energy; it can jump between shells by absorbing/releasing energy (any two) [2]. Rutherford: a revolving charged electron should lose energy and spiral into the nucleus, so the atom would be unstable — but atoms are stable [1].

Q19 **Question:** What would happen to the number of chromosomes over generations if gametes were formed by mitosis instead of meiosis? Explain. **[3]**

Expected answer / marking points: Mitosis keeps the chromosome number the same as the parent cell [1]. If gametes had the full number, the zygote formed by fusion would have double the chromosomes [1], and the number would keep doubling every generation. Meiosis halves the number in gametes so the species keeps a constant chromosome number [1].

Q20 **Question:**
[4]

> A farmer, Deepa, could sell only one-fourth of her amla and lemon harvest. To stop the rest from spoiling, she made pickles, murabba and sharbat by adding large amounts of salt, sugar or jaggery to small pieces of fruit and juice, and sold them in glass bottles.

(i) Which scientific concept did Deepa use to preserve the produce? [1]
(ii) How do high concentrations of salt and sugar stop the growth of bacteria and fungi? [2]
(iii) Suggest one more food that is preserved in a similar way at home. [1]

Expected answer / marking points: (i) Osmosis (water moving out of cells into a concentrated solution) [1]. (ii) The salt/sugar solution is very concentrated, so water moves out of microbial cells by osmosis [1]; the microbes lose water, shrink and cannot grow or multiply, so food does not spoil [1]. (iii) Any: jam, murabba, salted fish, sugar syrup fruits, lemon pickle [1].

Q21 **Question:** Draw a neat labelled diagram of a plant cell (any six labels). Write three differences between a
[5] plant cell and an animal cell.

ATHVA / OR

Name the four basic types of animal tissues. For each, write one place where it is found and one function. Draw a labelled diagram of any one of them.

Expected answer / marking points: Diagram with any six correct labels — cell wall, cell membrane, nucleus, chloroplast, large central vacuole, mitochondria, Golgi apparatus, ER, cytoplasm [3]. Differences: plant cells have a cell wall, chloroplasts and a large central vacuole; animal cells lack a cell wall and chloroplasts and have small or no vacuoles (any three) [2].
OR: Epithelial — skin/lining of organs, protection/exchange; Connective — bone/cartilage/blood/tendon, support/binding/transport; Muscular — arms/heart/stomach wall, movement; Nervous — brain/spinal cord/nerves, carry messages [4 × 0.75 = 3]. Labelled diagram (e.g., neuron with cell body, dendrites, axon) [2].

Score: ___ / 40 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Wednesday, 28 October 2026 · 10 marks

Test ID: D2026-10-28

SST Ch 4 Early humans · English Modals · AI Data Literacy · Maths 10.2 Data & graphs · Maths Ch 10 exercises

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Q1 [1]

Question: In which state are the Bhimbetka rock shelters located?

Expected answer / marking points: Madhya Pradesh.

Q2 [2]

Question: Write two important changes in human life during the Neolithic period.

Expected answer / marking points: Farming and domestication; permanent villages; pottery; weaving; polished tools (any two) [1 + 1].

Q3 [2]

Question: Fill in with suitable modals: (i) You _____ not use your phone during the exam. (ii) _____ I borrow your pen, please?

Expected answer / marking points: (i) must/should [1] (ii) May/Can [1]

Q4 [2]

Question: (i) Classify as qualitative or quantitative data: height of students; favourite colour. (ii) Give one example of structured data.

Expected answer / marking points: (i) Height — quantitative; favourite colour — qualitative [1] (ii) A table/spreadsheet such as a marks list or attendance register [1].

Q5 [1]

Question: Which type of graph is best for showing how a quantity changes over time?

Expected answer / marking points: Line graph.

Q6 [2]

Question: Shreyas holds 25 shares at ₹150 average and buys 10 more at ₹30. Find his new average price per share.

Expected answer / marking points: Total = 3750 + 300 = 4050 [1]; ÷ 35 ≈ ₹115.71 [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Thursday, 29 October 2026 · 10 marks

Test ID: D2026-10-29

Hindi पाठ 8 रैदास · Maths Ch 3 World of Numbers · SST Ch 5 State & Society · Maths Ch 10 exercises

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Q1 [1]

Question: संत रैदास का जन्म कहाँ हुआ था?

Expected answer / marking points: काशी (वाराणसी) में।

Q2 [2]

Question: "प्रभु जी तुम दीपक, हम बाती" — इस पंक्ति का भाव लिखिए।

Expected answer / marking points: दीपक-बाती मिलकर ज्योति जलाते हैं, वैसे ही भक्त प्रभु से जुड़कर जीवन आलोकित करता है; अटूट संबंध [2]।

Q3 [1]

Question: Is 0.101001000100001... rational or irrational? Why?

Expected answer / marking points: Irrational — its decimal expansion is non-terminating and non-repeating.

Q4 [2]

Question: Find three rational numbers between 1/6 and 2/5.

Expected answer / marking points: 1/6 = 5/30, 2/5 = 12/30 ⇒ e.g., 6/30 (= 1/5), 7/30, 8/30 (= 4/15) [method 1 + answer 1].

Q5 [1]

Question: What is meant by the idea of a *chakravarti samrāṭ*?

Expected answer / marking points: The ideal of a pan-Indian (universal) monarch whose authority extends over the subcontinent.

Q6 [1]

Question: Which socio-ethical principle continued as a binding idea throughout the period up to 1000 CE?

Expected answer / marking points: Dharma.

Q7 [2]

Question: Shreyas holds 25 shares at ₹150 average and buys 10 more at ₹30. Find his new average price per share.

Expected answer / marking points: Total = 3750 + 300 = 4050 [1]; ÷ 35 ≈ ₹115.71 [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Friday, 30 October 2026 · 10 marks

Test ID: D2026-10-30

Science Ch 3 Plant tissues · Hindi अलंकार · English Unit 5 (poem) · Science 8.7 Electron shells

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Q1 **[2]**

Question: Write two features of meristematic cells. Name any two types of meristem by location.

Expected answer / marking points: Thin walls, dense cytoplasm, large nucleus, divide repeatedly (any two) [1]; apical, lateral, intercalary (any two) [1].

Q2 **[1]**

Question: Which plant tissue gives coconut husk its toughness?

Expected answer / marking points: Sclerenchyma.

Q3 **[3]**

Question: अलंकार पहचानिए — (क) रघुपति राघव राजा राम (ख) काली घटा का घमंड घटा (ग) मंगन को देखि पट देत बार-बार है

Expected answer / marking points: (क) अनुप्रास (ख) यमक — "घटा": बादल / कम हुआ (ग) श्लेष — "पट": वस्त्र / दरवाज़ा [1 × 3]

Q4 **[2]**

Question: In the Unit 5 poem, why did the other runners turn back during the race? What quality does this show?

Expected answer / marking points: To help the young boy who had fallen [1]; kindness, empathy, sportsmanship [1].

Q5 **[1]**

Question: What is the maximum number of electrons the L-shell can hold?

Expected answer / marking points: 8 ($2n^2$ with n = 2).

Q6 **[1]**

Question: Write the electronic configuration of sodium (atomic number 11).

Expected answer / marking points: 2, 8, 1

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Saturday, 31 October 2026 · 10 marks

Test ID: D2026-10-31

SST Ch 4 Early humans · Maths Ch 3 World of Numbers · Hindi पाठ 5 · Maths 11.1 Algorithms

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Q1 [1]

Question: Name the site in the north-west where early farming began around 7000 BCE.

Expected answer / marking points: Mehrgarh.

Q2 [2]

Question: Give one advantage and one challenge for early people who settled near rivers.

Expected answer / marking points: Advantage: fertile soil/water/transport [1]; challenge: floods/changing course/disease [1].

Q3 [1]

Question: Without dividing, say whether 18/125 has a terminating decimal expansion. Why?

Expected answer / marking points: Yes — the denominator 125 = 5^3 has no prime factor other than 2 or 5.

Q4 [2]

Question: Express 0.363636... in the form p/q.

Expected answer / marking points: x = 0.3636..., 100x = 36.36... ⇒ 99x = 36 [1] ⇒ x = 36/99 = 4/11 [1].

Q5 [1]

Question: "आखिरी चट्टान तक" किस विधा की रचना है?

Expected answer / marking points: यात्रा-वृत्तांत।

Q6 [1]

Question: What is an algorithm?

Expected answer / marking points: A finite, step-by-step list of clear instructions to solve a problem or do a task.

Q7 [2]

Question: Write an algorithm (steps) to find the sum 1 + 2 + ... + n, and use it to find the sum for n = 10.

Expected answer / marking points: Steps: set sum = 0; for k = 1 to n add k to sum; output sum [1]. For n = 10, sum = 55 (or n(n + 1)/2) [1].

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Daily Test · Answer key

Sunday, 1 November 2026 · 10 marks

Test ID: D2026-11-01

English Unit 3 · AI Communication Skills · SST Ch 5 State & Society · Science 8.8 Valency

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Q1 [1]

Question: In “Winds of Change”, what is a *pankha*?

Expected answer / marking points: A traditional hand fan.

Q2 [1]

Question: Why did pankhas from different regions of India look different from one another?

Expected answer / marking points: Each place used local materials (sola, bamboo, beads, leather) and its own designs.

Q3 [2]

Question: Name two barriers to effective communication and one way to overcome each.

Expected answer / marking points: Any two: language — use simple words; noise — move to a quiet place; emotional — stay calm, listen; cultural — be respectful [1 + 1].

Q4 [2]

Question: Differentiate between verbal and non-verbal communication with one example of each.

Expected answer / marking points: Verbal uses words — spoken or written (phone call, letter) [1]; non-verbal uses gestures, expressions, posture, eye contact (smile, thumbs-up) [1].

Q5 [1]

Question: What were mahājanapadas?

Expected answer / marking points: Large, territory-based kingdoms that grew from the earlier janapadas.

Q6 [1]

Question: Name the ritual through which early kings proclaimed their supreme power.

Expected answer / marking points: The aśhvamedha yajña.

Q7 [1]

Question: What is meant by the valency of an element?

Expected answer / marking points: Its combining capacity — the number of hydrogen (or chlorine) atoms one atom can combine with.

Q8 [1]

Question: What is the valency of magnesium (2, 8, 2)?

Expected answer / marking points: 2

Score: ___ / 10 Checked by: _____

Weak topic aur correction:

Parent Sunday handover checklist

- [ ] Agle Monday–Sunday ka student pack print karke diya.
- [ ] Parent solved key alag rakhi; student copy ke saath share nahi ki.
- [ ] Pichhle week ka pending kaam aur 3 weak topics review kiye.
- [ ] School exam date sheet aur tuition timings se schedule match kiya.

Agli priority / teacher se poochhna: